

Etapas y render graphs

Y cómo funcionan en Unity

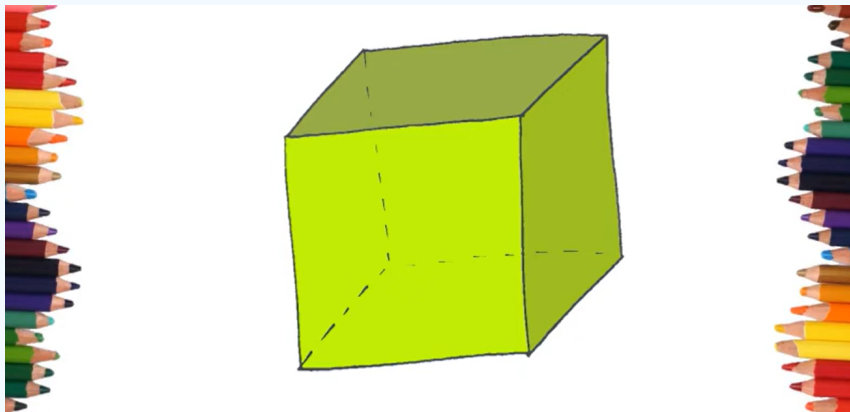
Marco Antonio Gómez Martín
Iluminación y Materiales
Grado en Desarrollo de Videojuegos
2025/2026



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Frame buffer

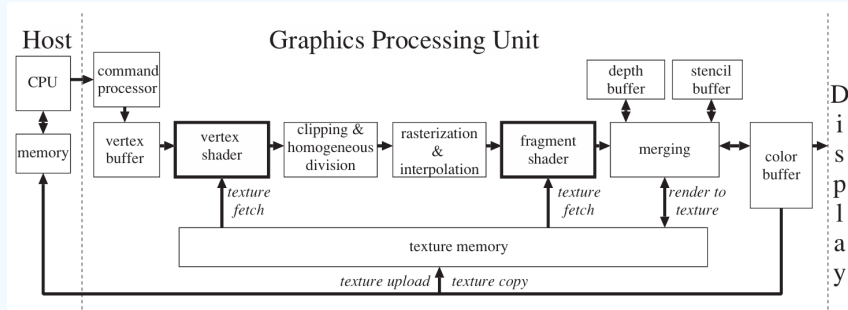
- Función `display()` de la ventana con doble buffer: **Front** y **Back**:

```
// El BackColorBuffer queda del color fijado
// con glClearColor()
// El DepthBuffer queda a 1 (máxima distancia)
glClear(GL_COLOR_BUFFER_BIT | GL_DEPTH_BUFFER_BIT);

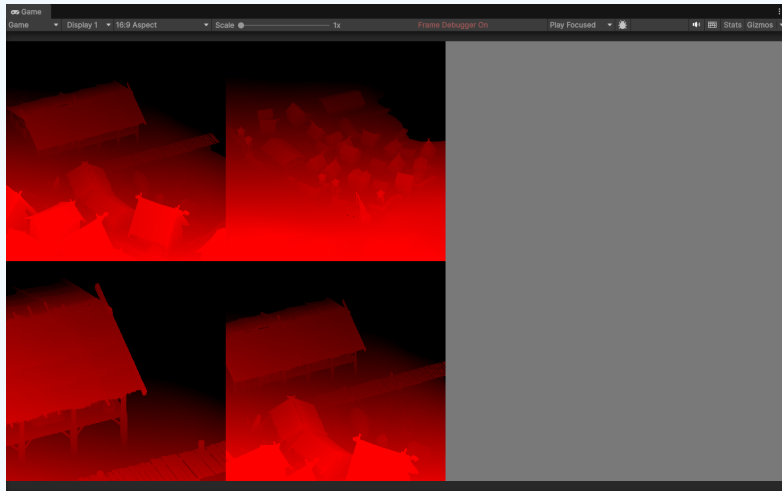
// dibuja los objetos en el BackColorBuffer
scene.render();

// intercambia los buffers (Back/Front)
glfwSwapBuffers(mWindow);
```

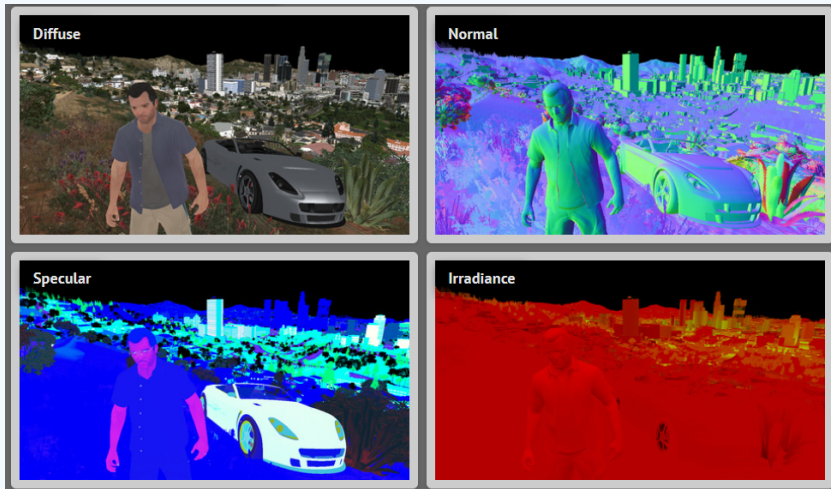
Render Targets



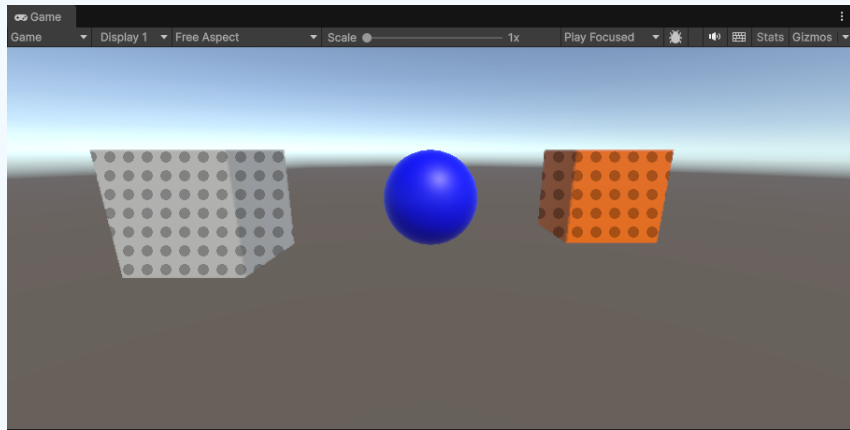
Render Targets

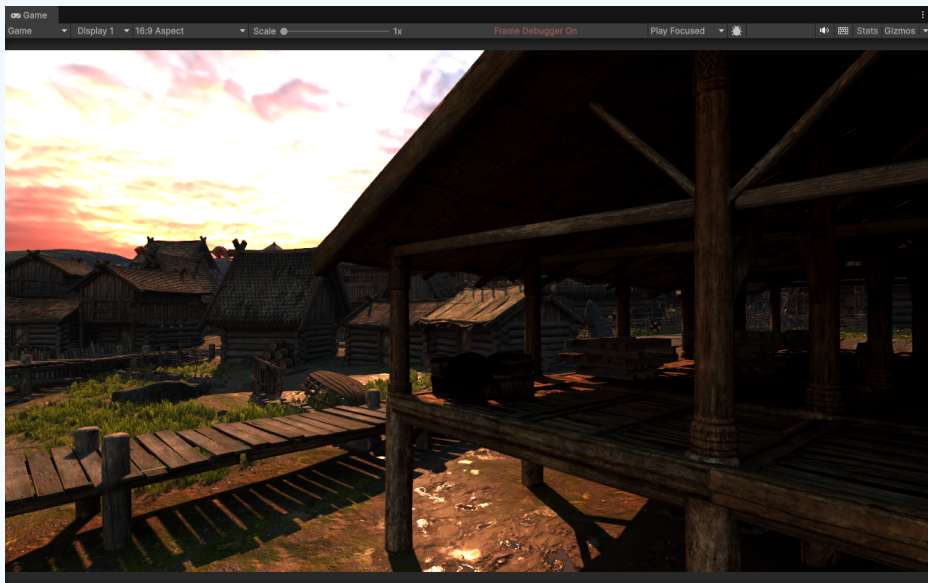


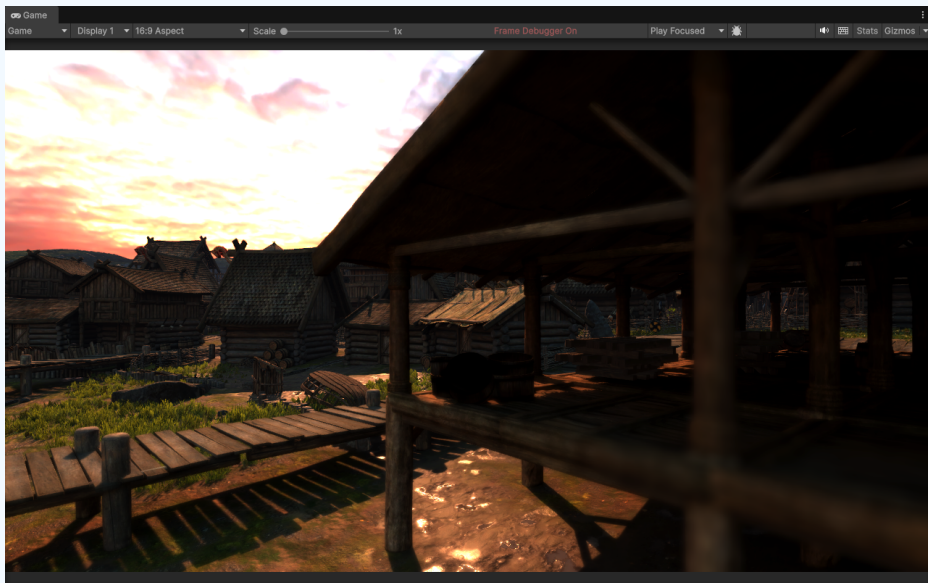
Render Targets

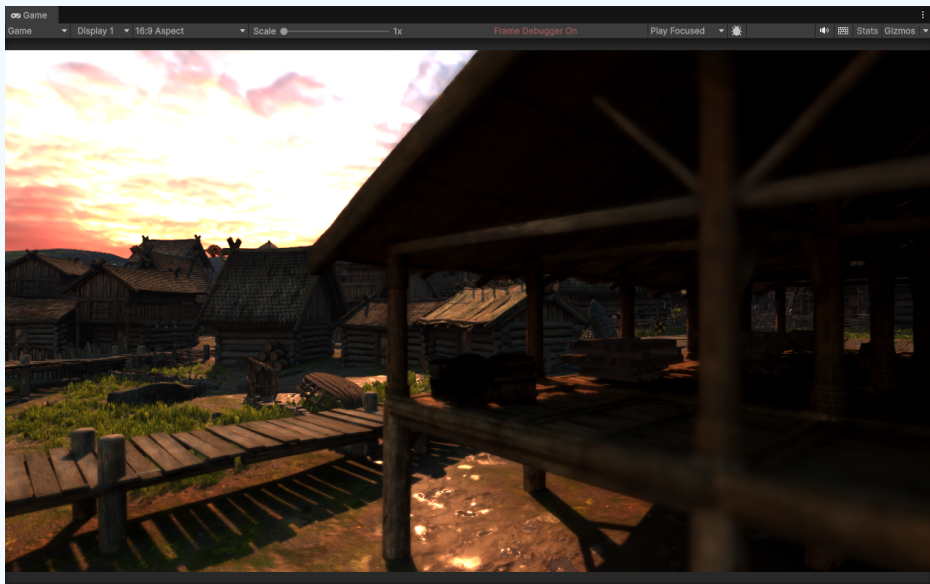


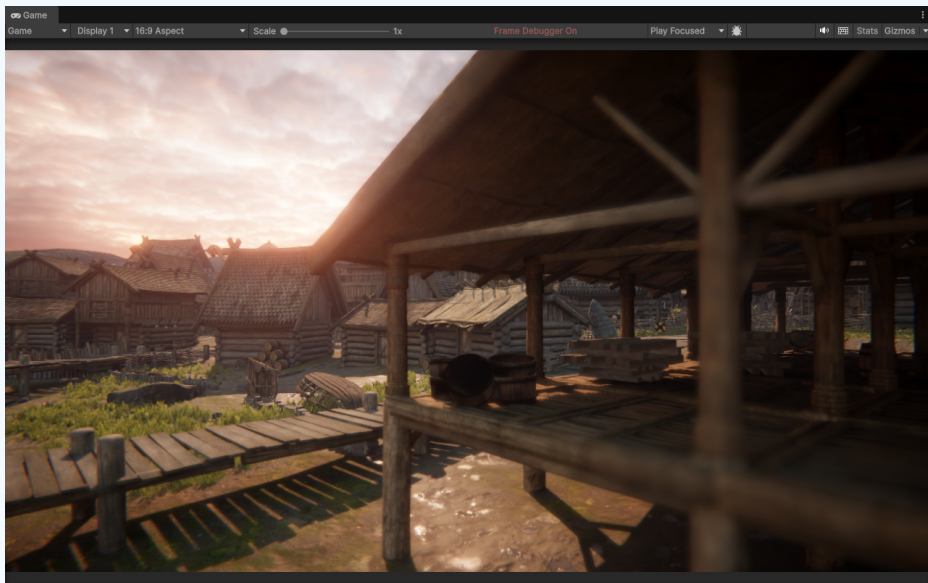
Render Targets



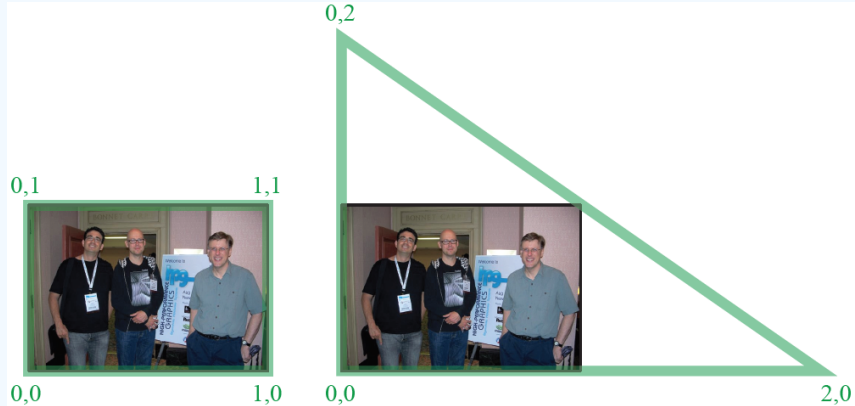






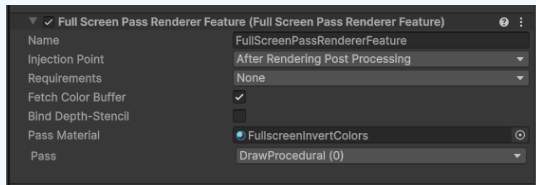


Render Targets - Postprocesado



Postprocesado simple en URP

- Render feature disponible en Unity: *Full Screen Pass Renderer Feature*
- Permite configurar el shader/material que se utilizará y de él, qué pasada.
- La textura con el contenido se envía al shader en el parámetro `_BlitTexture`.



Postprocesado simple en URP

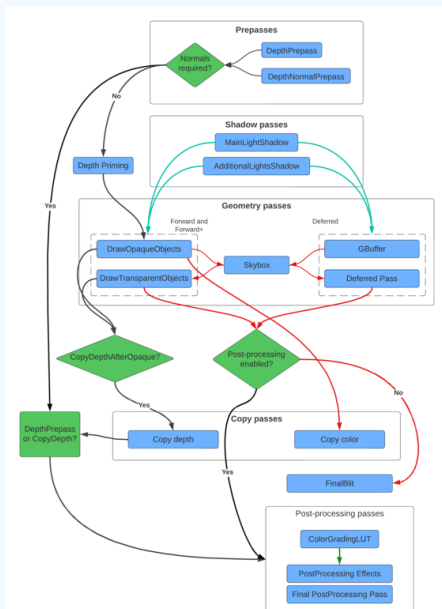
Esqueleto de shader de postprocesado

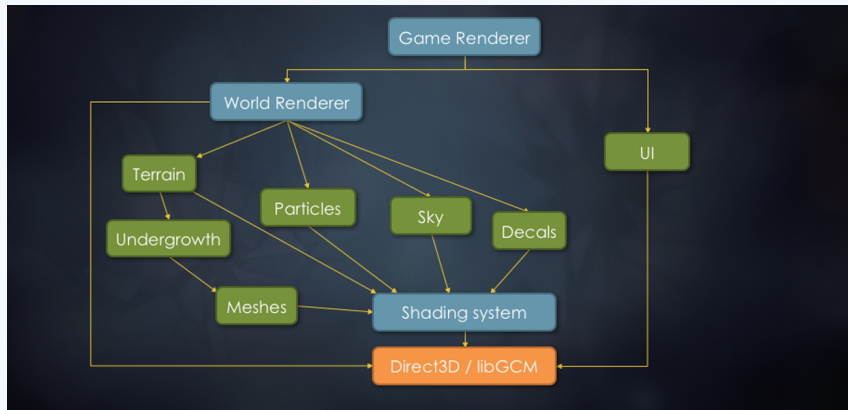
```
SubShader {  
    Cull Off ZWrite Off ZTest Always // No culling or depth  
    Pass {  
        Name "Negative"  
  
        HLSLPROGRAM  
  
        #include "Packages/com.unity.render-pipelines.universal/\n                ShaderLibrary/Core.hlsl"  
        #include "Packages/com.unity.render-pipelines.core/Runtime/\n                Utilities/Blit.hlsl"  
  
        ...  
  
        ENDHLSL  
    }  
}
```

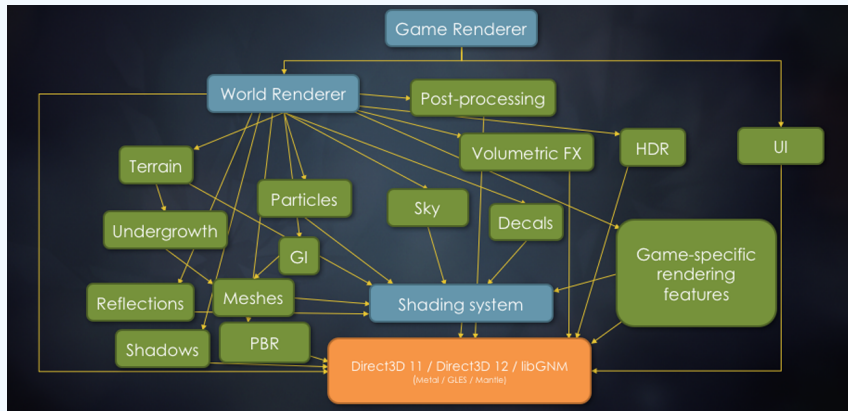
Postprocesado simple en URP

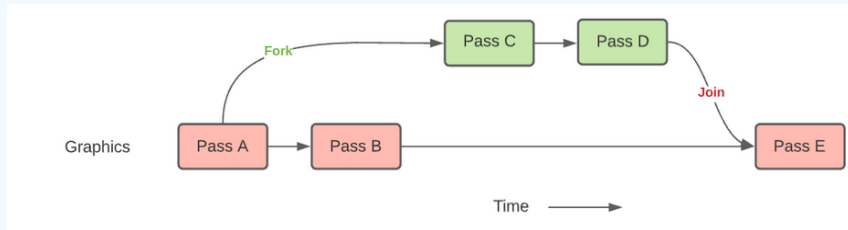
Ejemplo

```
...  
#pragma vertex Vert // Funcion Vert en Blit.hlsl  
#pragma fragment frag  
  
//TEXTURE2D_X(_BlitTexture);  
SAMPLER(sampler_BlitTexture);  
  
half4 frag (Varyings input) : SV_Target  
{  
    return 1 - SAMPLE_TEXTURE2D_X(_BlitTexture,  
                                   sampler_BlitTexture, input.texcoord);  
}  
...
```











GDC

FrameGraph: Extensible Rendering Architecture in Frostbite

Yuriy O'Donnell
Rendering Engineer
Frostbite

GAME DEVELOPERS CONFERENCE[®] | FEB 27-MAR 3, 2017 | EXPO: MAR 1-3, 2017 #GDC17



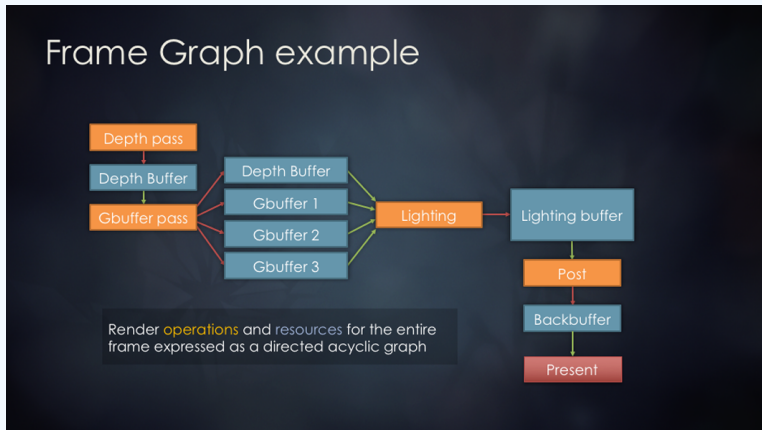
[https://www.gdcvault.com/play/1024612/
FrameGraph-Extensible-Rendering-Architecture-in](https://www.gdcvault.com/play/1024612/FrameGraph-Extensible-Rendering-Architecture-in)

Render graphs

Distintos nombres

- Frame graph (GCD 2017)
- Render graphs (Unity)
- Render dependency graphs (Unreal Engine)

Render graphs

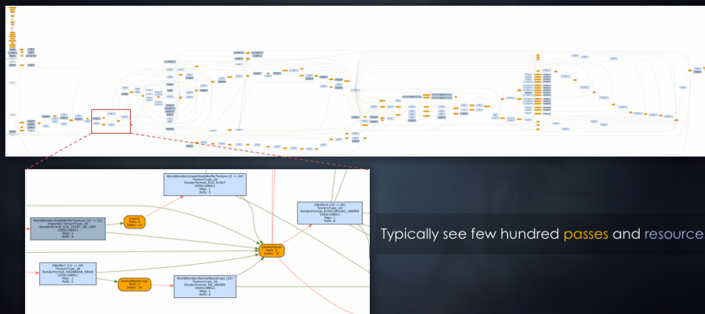


<https://www.gdcvault.com/play/1024612/>

FrameGraph-Extensible-Rendering-Architecture-in

Render graphs

Graph of a Battlefield 4 frame

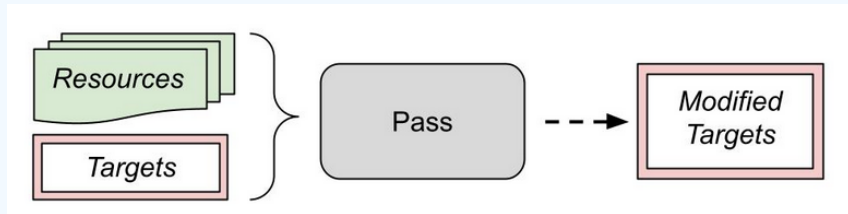


Typically see few hundred **passes** and resources

<https://www.gdcvault.com/play/1024612/FrameGraph-Extensible-Rendering-Architecture-in>

Render graphs - Pasadas

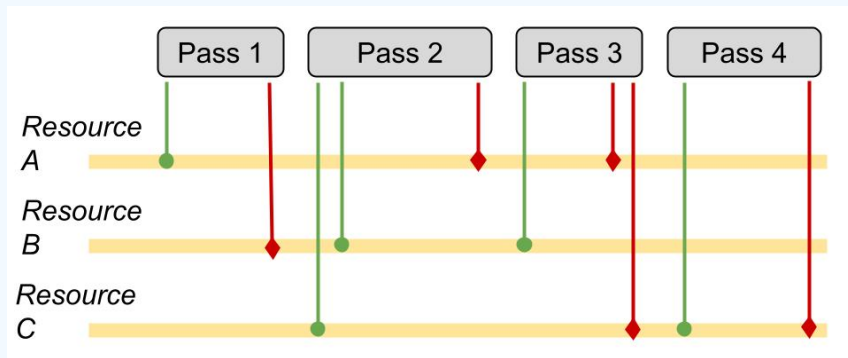
El renderizado se divide en pasadas



Render graphs - Fases

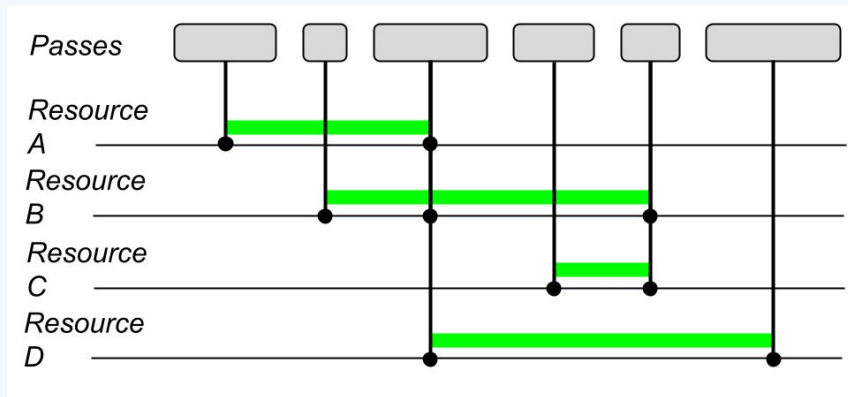
- **Setup:** se declaran las pasadas y los recursos de entrada y salida que utilizan. Los recursos son "virtuales".
- **Compile:** se determina el ciclo de vida de los recursos, se eliminan pasadas no relevantes, se hacen las reservas de recursos necesarias.
- **Execute:** se ejecutan las pasadas.

Render graphs - Ventajas



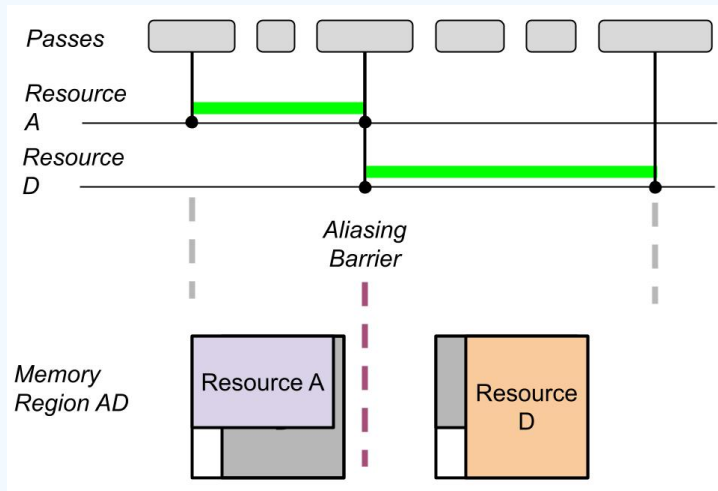
<https://logins.github.io/graphics/2021/05/31/RenderGraphs.html>

Render graphs - Ventajas



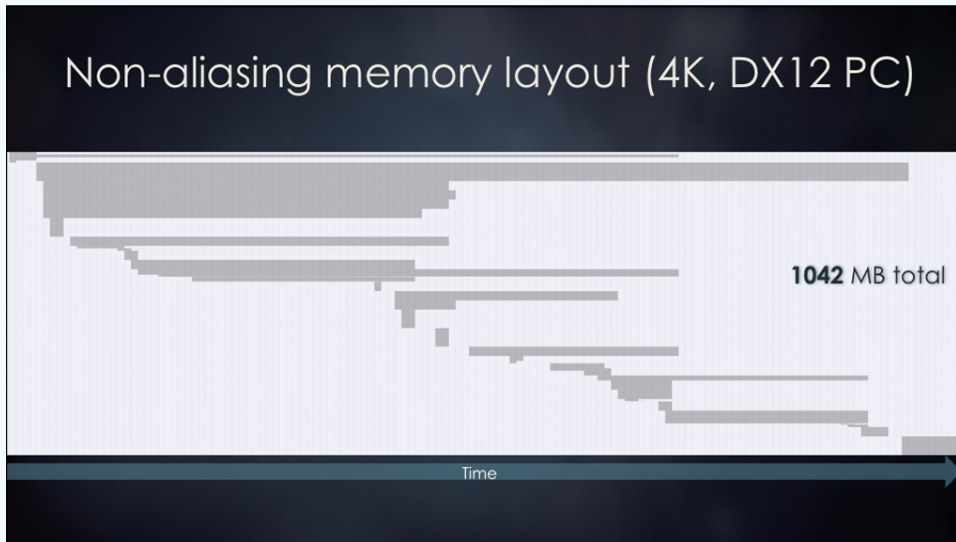
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Render graphs - Ventajas



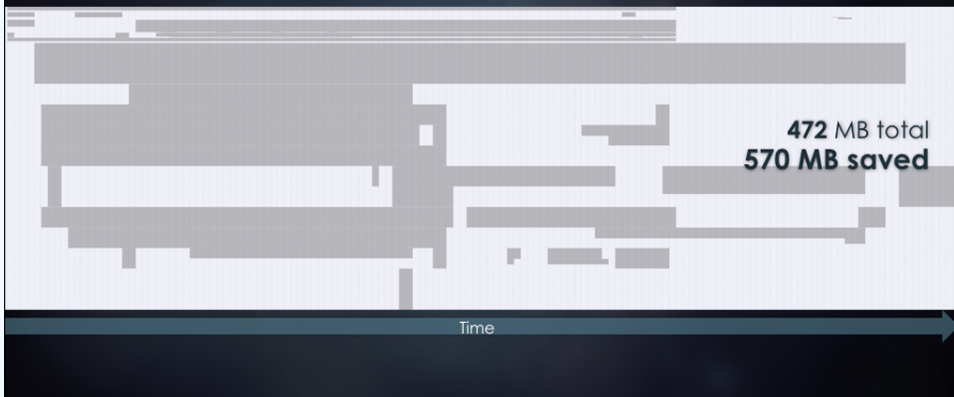
<https://logins.github.io/graphics/2021/05/31/RenderGraphs.html>

Render graphs - Ventajas



Render graphs - Ventajas

Aliasing memory layout (4K, DX12 PC)





Scriptable Render Pipeline

- Unity incluye la implementación de una render graph con API en C#.
- Si se quiere construir un render pipeline nuevo, se utiliza en cada frame para construir las etapas que hay que hacer.
- Unity es responsable de compilar y ejecutar el grafo que resulte.

SRP Render pipeline a vista de pájaro

```
using System;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.Rendering;
using UnityEngine.Rendering.RenderGraphModule;

public class ExampleRenderPipeline : RenderPipeline {

    private RenderGraph myRenderGraph;

    public ExampleRenderPipeline(ExampleRenderPipelineAsset asset)
    {
        myRenderGraph = new RenderGraph("My render graph");
    }

    ...
}
```

SRP Render pipeline a vista de pájaro

```
public class ExampleRenderPipeline : RenderPipeline {  
    ...  
  
    protected override void Render(ScriptableRenderContext context,  
                                    List<Camera> cameras) {  
        foreach (var cam in cameras)  
        {  
            if (cam.enabled)  
            {  
                RenderCamera(context, cam);  
            }  
        }  
    }  
  
    ...  
}
```

SRP Render pipeline a vista de pájaro

```
void RenderCamera(ScriptableRenderContext context,
                  Camera cameraToRender) {

    context.SetupCameraProperties(cameraToRender);
    var cmd = CommandBufferPool.Get("Example command buffer");

    RenderGraphParameters rgParams = new RenderGraphParameters {
        commandBuffer = cmd,
        scriptableRenderContext = context,
        currentFrameIndex = Time.frameCount,
    };

    ...

    context.ExecuteCommandBuffer(cmd);
    CommandBufferPool.Release(cmd);
    context.Submit();
}
```

SRP Render pipeline a vista de pájaro

```
void RenderCamera(ScriptableRenderContext context,
                  Camera cameraToRender) {

    ...

    try
    {
        myRenderGraph.BeginRecording(rgParams);
        // Add render passes here.
        myRenderGraph.EndRecordingAndExecute();
    }
    catch (Exception e)
    {
        if (renderGraph.ResetGraphAndLogException(e))
            throw;
    }

    ...

}
```


SRP Render pipeline a vista de pájaro

- `// Add render passes here` significa "añadir nodos al render graph".
- Los nodos se añaden en el orden en el que se ejecutarán.

¿Y en URP cómo es?

Etapas y render graphs

Y cómo funcionan en Unity

¿Preguntas?



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